



Modeling air pollution exposure with MATSim: to what extent are complex modeling chains necessary?

Marjolaine Lannes^{1,2}, Nicolas Coulombel², Yelva Roustan¹

¹ CEREIA, ENPC, Institut Polytechnique de Paris, EDF R&D, Marne-la-Vallée, France

² LVMT, ENPC, Institut Polytechnique de Paris, Univ. Eiffel, Marne-la-Vallée, France

MATSim User Meeting 2026

Evaluating health impacts of air pollution and policy responses

Ambient air pollution is a major **health concern**:

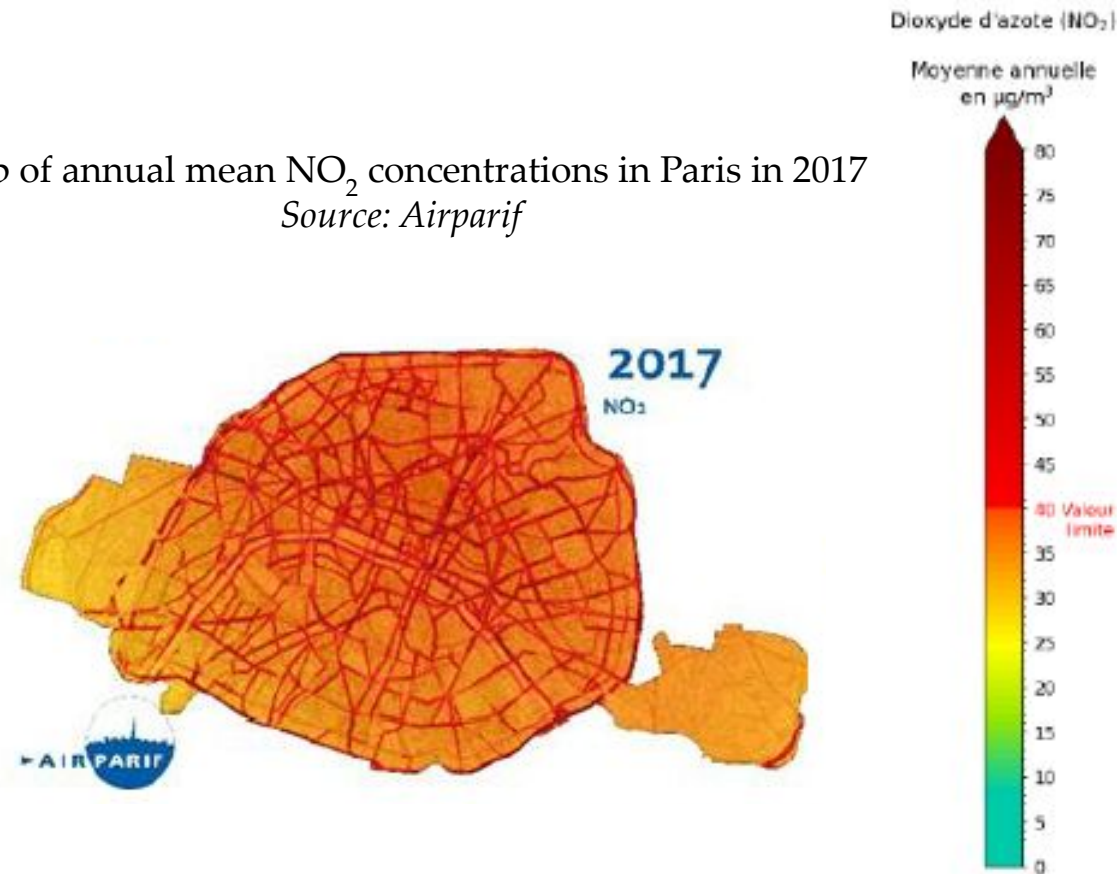
- 4.2 million premature deaths/year worldwide [1]
- Regulated pollutants: fine particles (PM_{2.5}), nitrogen dioxide (NO₂), ozone (O₃)

In urban areas, important **heterogeneities** in the concentrations of pollutants at the local scale

Transportation plays a key role in emissions and public policies aimed at mitigating health impacts

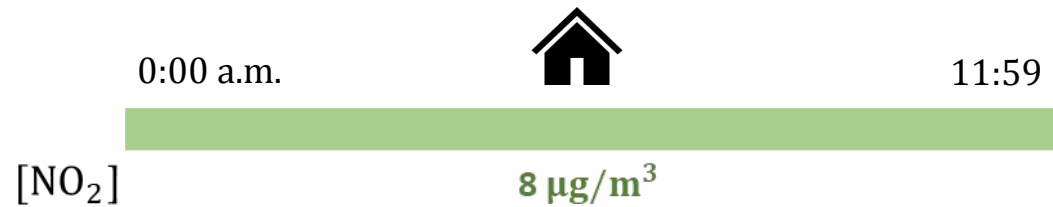
Map of annual mean NO₂ concentrations in Paris in 2017

Source: Airparif



Modeling issues of air pollution exposure

- Reference method: **static** approach (residential locations) with **background** concentrations

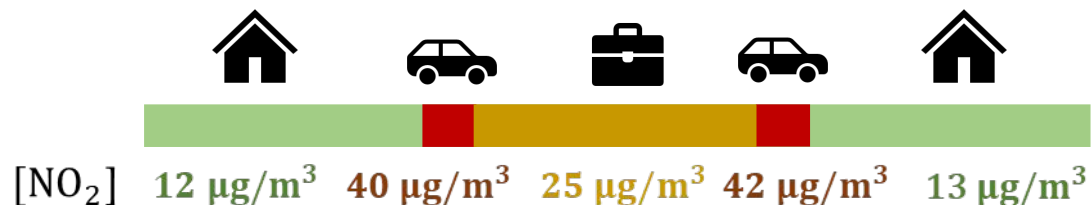


Modeling issues of air pollution exposure

- Reference method: **static** approach (residential locations) with **background** concentrations



- Recent work with a **dynamic** approach
 - exposure during activities and travel
 - exposure inequalities with ABMs

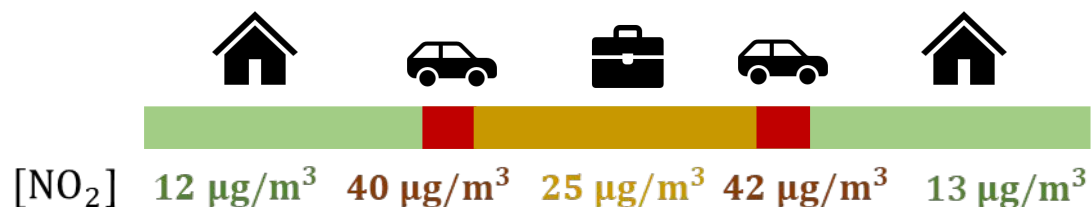


Modeling issues of air pollution exposure

- Reference method: **static** approach (residential locations) with **background** concentrations



- Recent work with a **dynamic** approach
 - exposure during activities and travel
 - exposure inequalities with ABMs



Development of **modeling chains** to assess individual exposure coupling **mobility ABMs** and **air quality** models to ensure consistency [2] [3]

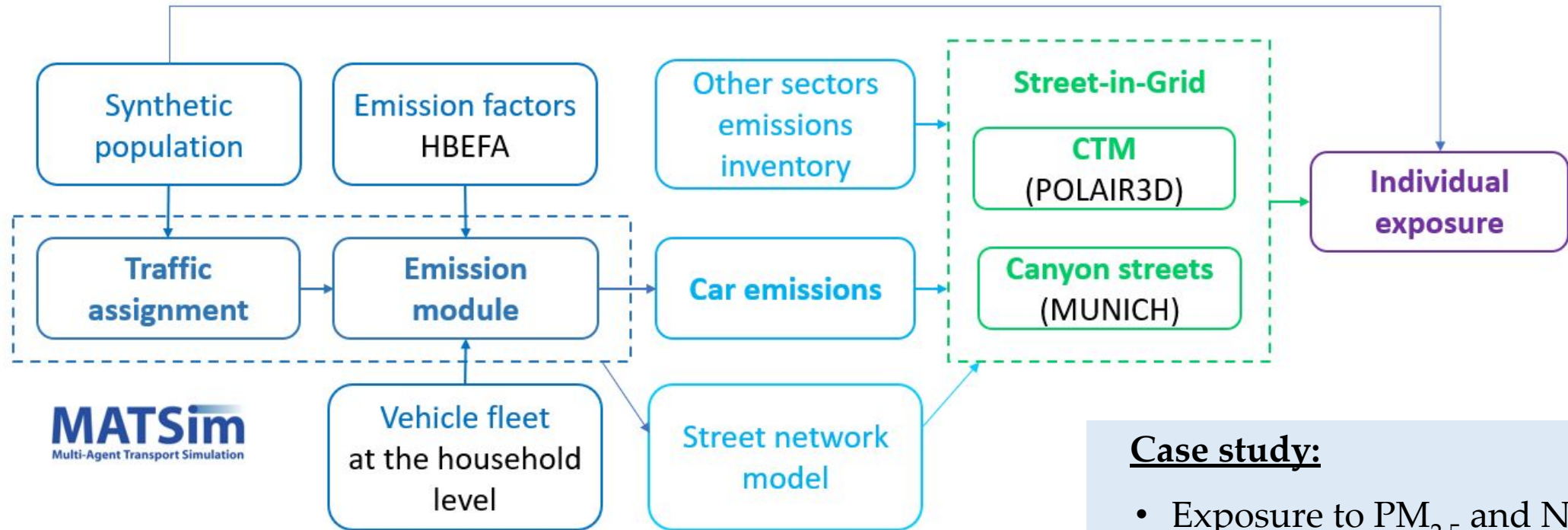
Increasing complexity: higher resolution and integration of additional components

- **To what extent is complexity warranted in modeling individual exposure to ambient air pollution in urban areas?**

[2] Gastineau et al. 2023 ;

[3] Gurram, Sashikanth, Amy Lynette Stuart, et Abdul Rawoof Pinjari. 2019.

Modeling chain and case study



Population and travel demand: eqasim [4]

Traffic assignment: MATSim [5]

Emissions: HBEFA [6] (average / detailed car emissions)

Case study:

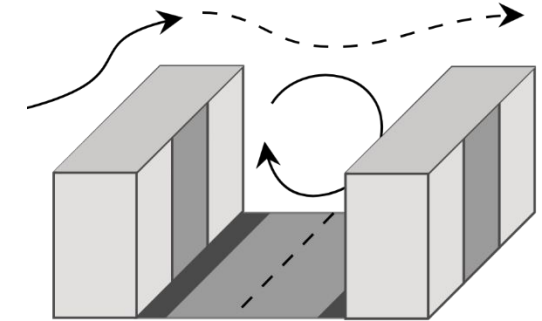
- Exposure to PM_{2.5} and NO₂
- Paris region (Île-de-France)
- Population sample size 25% (3 million inhabitants)
- Winter 2018

[4] Hörl and Balac, 2021

[5] Axhausen, Horni, and Nagel, 2016 ; [6] Kickhöfer, 2016

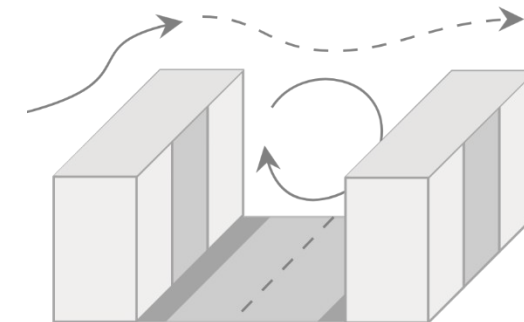
Multi-scale air quality modeling

- **Canyon street:** strong influence of the aerodynamic effect of buildings [7]
 - underestimation of concentrations in densely built areas by regional chemistry-transport models (CTMs)



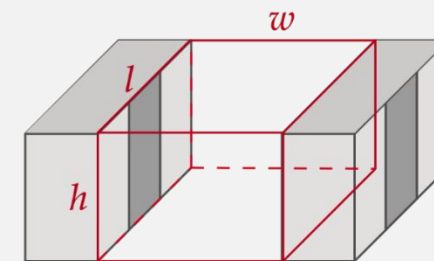
Multi-scale air quality modeling

- **Canyon street:** strong influence of the aerodynamic effect of buildings [9]
 - underestimation of concentrations in densely built areas by regional chemistry-transport models (CTMs)

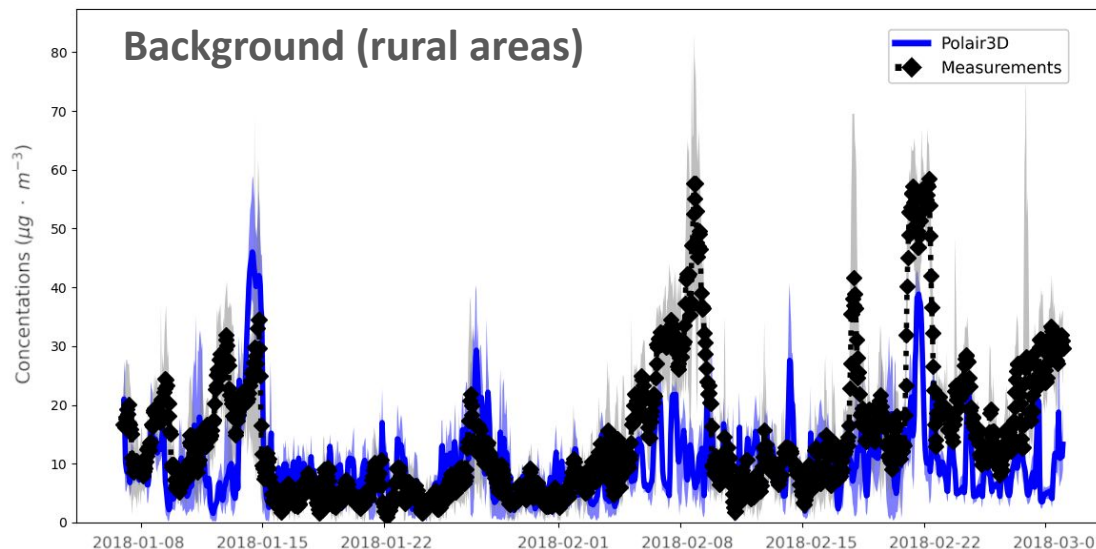
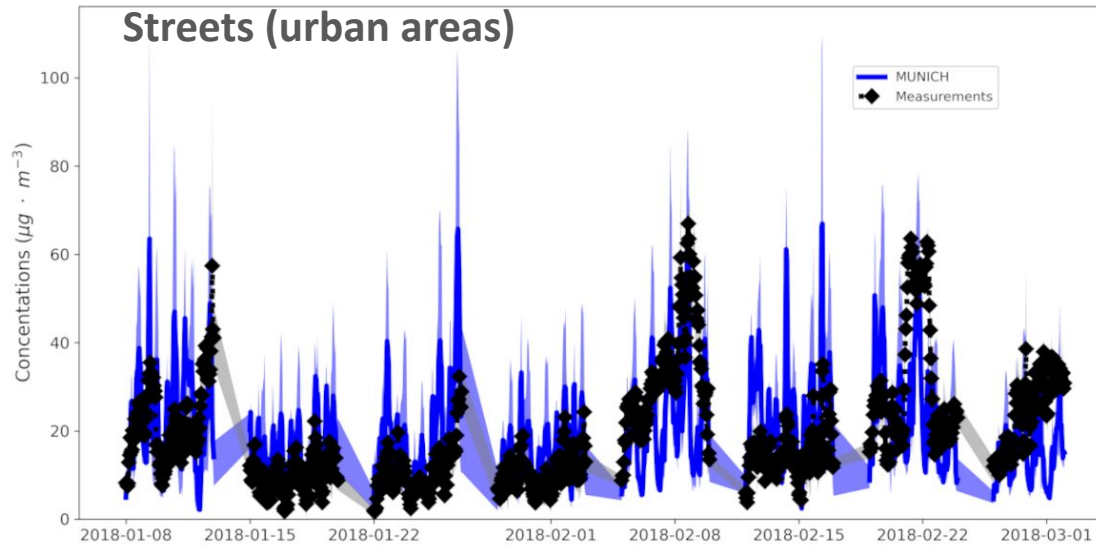


Polair3D & MUNICH

- **Background** concentrations: Eulerian CTM (Polair3D) [8]
- Concentrations in **canyon streets**: street box model (MUNICH) [9]
 - Both able to represent secondary pollutants
- Main input data: meteorological conditions, land use, emissions, street network
- **Exposure modeling:**
 - Buildings in wide streets: regional scale model
 - Buildings along canyon streets: local scale model

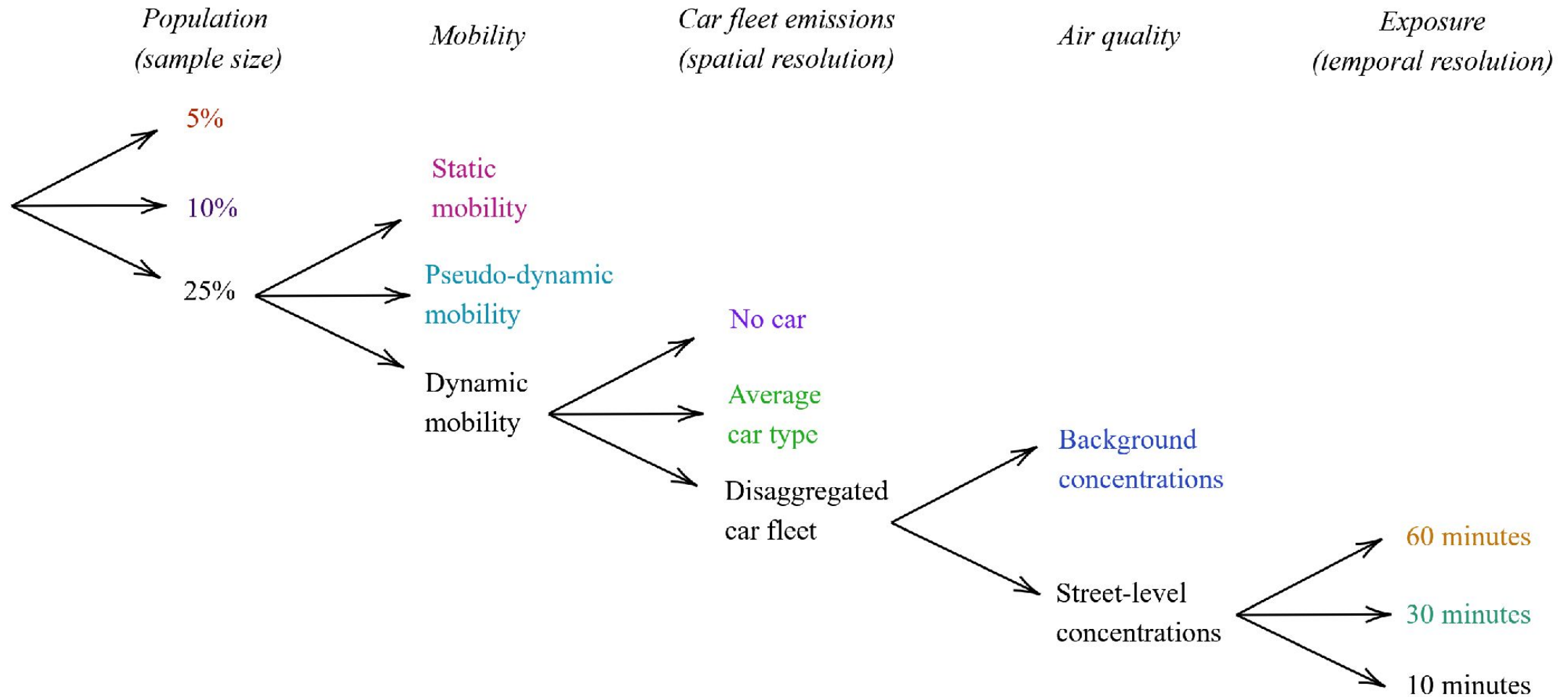


Assessment of simulated fine particle concentrations (PM_{2.5})

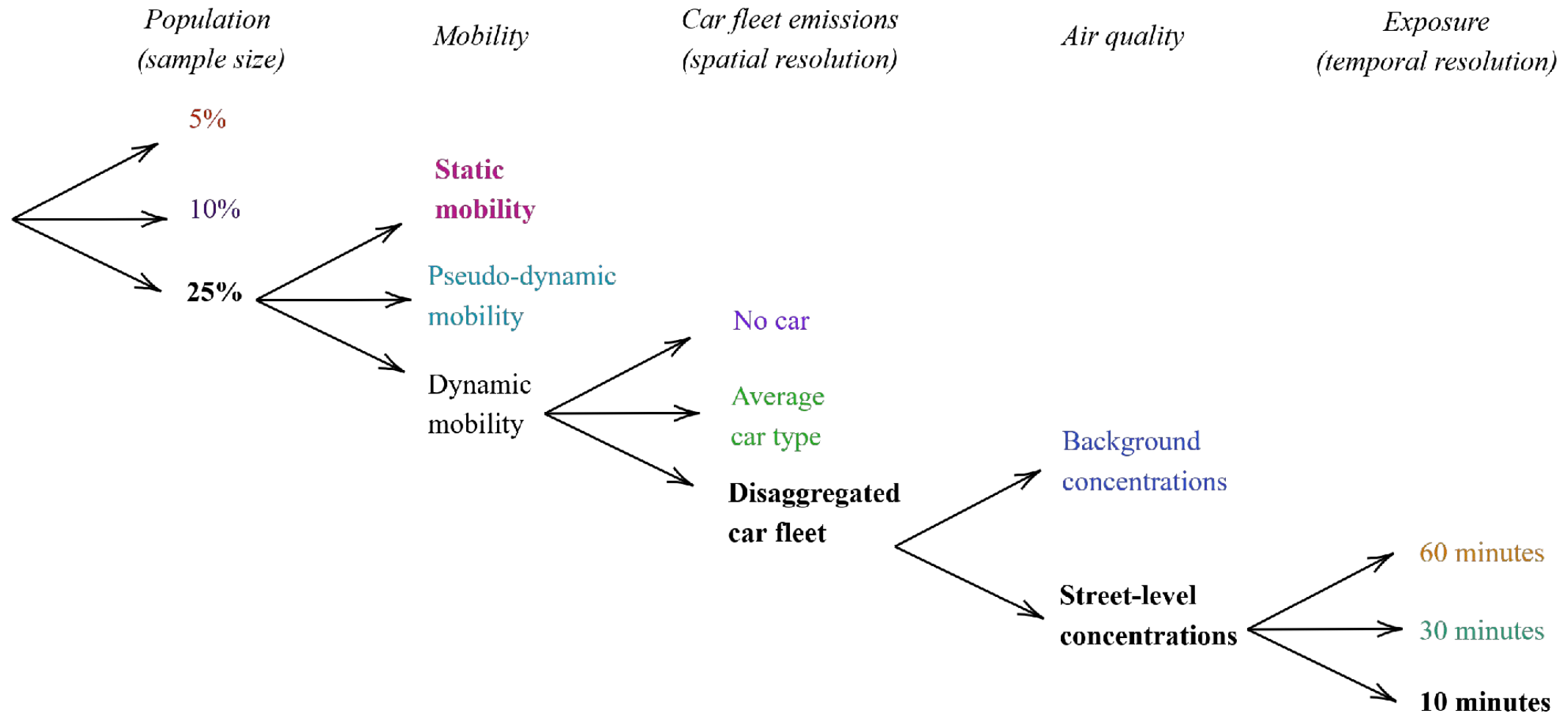


	Polair3D			MUNICH		
	NO ₂	PM _{2.5}	PM ₁₀	NO ₂	PM _{2.5}	PM ₁₀
Number of stations	27	9	14	8	3	6
Obs. mean ($\mu\text{g}/\text{m}^3$)	32.7	14.8	20.1	57.7	19.8	31.8
Sim. mean ($\mu\text{g}/\text{m}^3$)	21.7	9.6	18.7	38.3	19.9	26.6
RMSE ($\mu\text{g}/\text{m}^3$)	17.4	13.4	12.6	30.3	16.1	18.7
Correlation (%)	64.1	36.4	51.6	50.4	37.6	46.4
NME (%)	42.2	60.5	44.7	42.4	58.6	42.1
NMB (%)	-35.1	-35.1	-7.4	-34.4	1.0	-17.8
MNBE (%)	-24.9	47.0	51.4	-29.1	40.9	4.8
MNGE (%)	41.4	106.5	83.1	41.1	79.0	48.8
FB (%)	-44.0	-43.0	-8.0	-43.0	-4.0	-20.0
FAC2 (%)	70.0	54.0	74.0	67.0	64.0	76.0
NMSE (%)	44.0	105.0	36.0	41.0	59.0	37.0

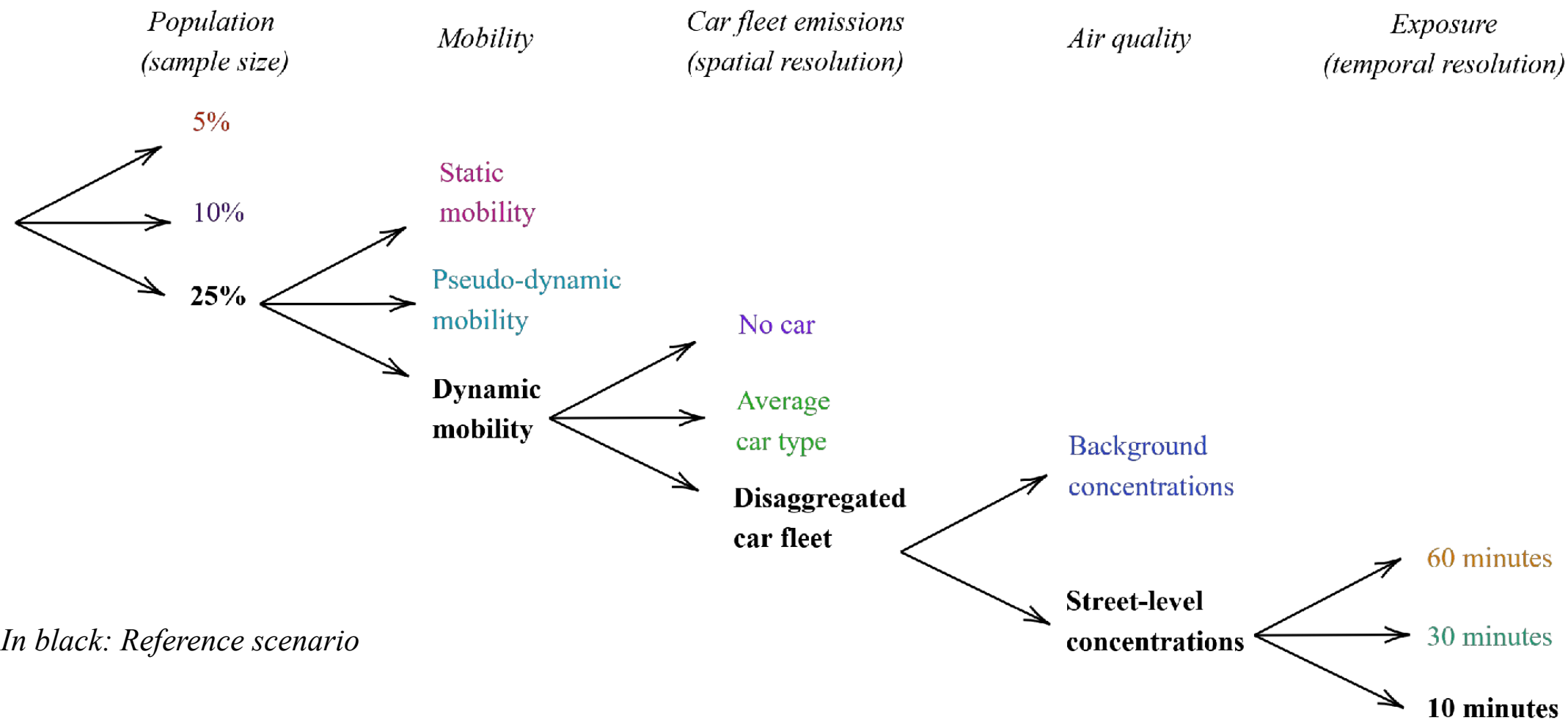
Definition of scenarios



Definition of scenarios



Definition of scenarios



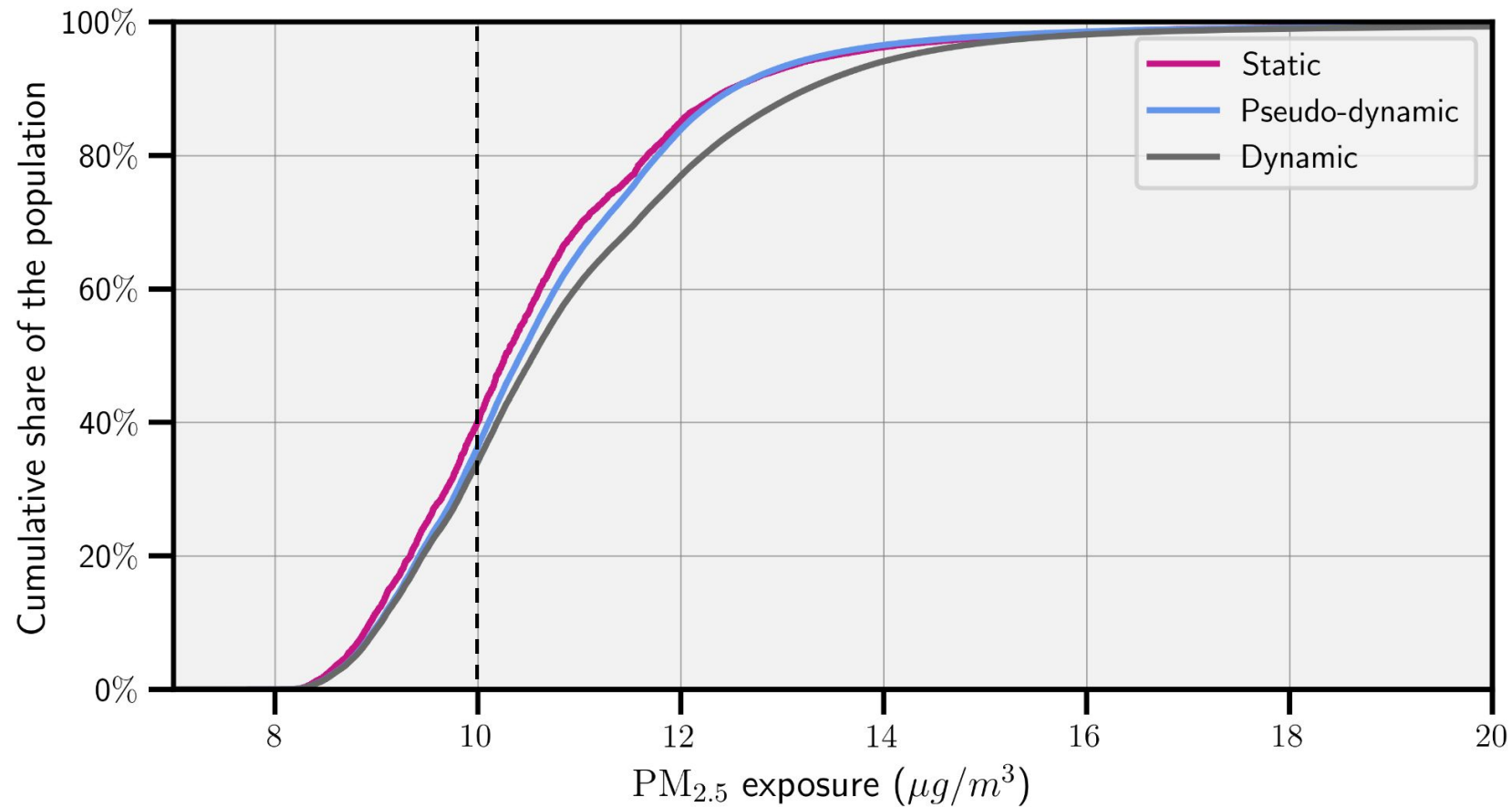
Analysis of scenarios

Indicators:

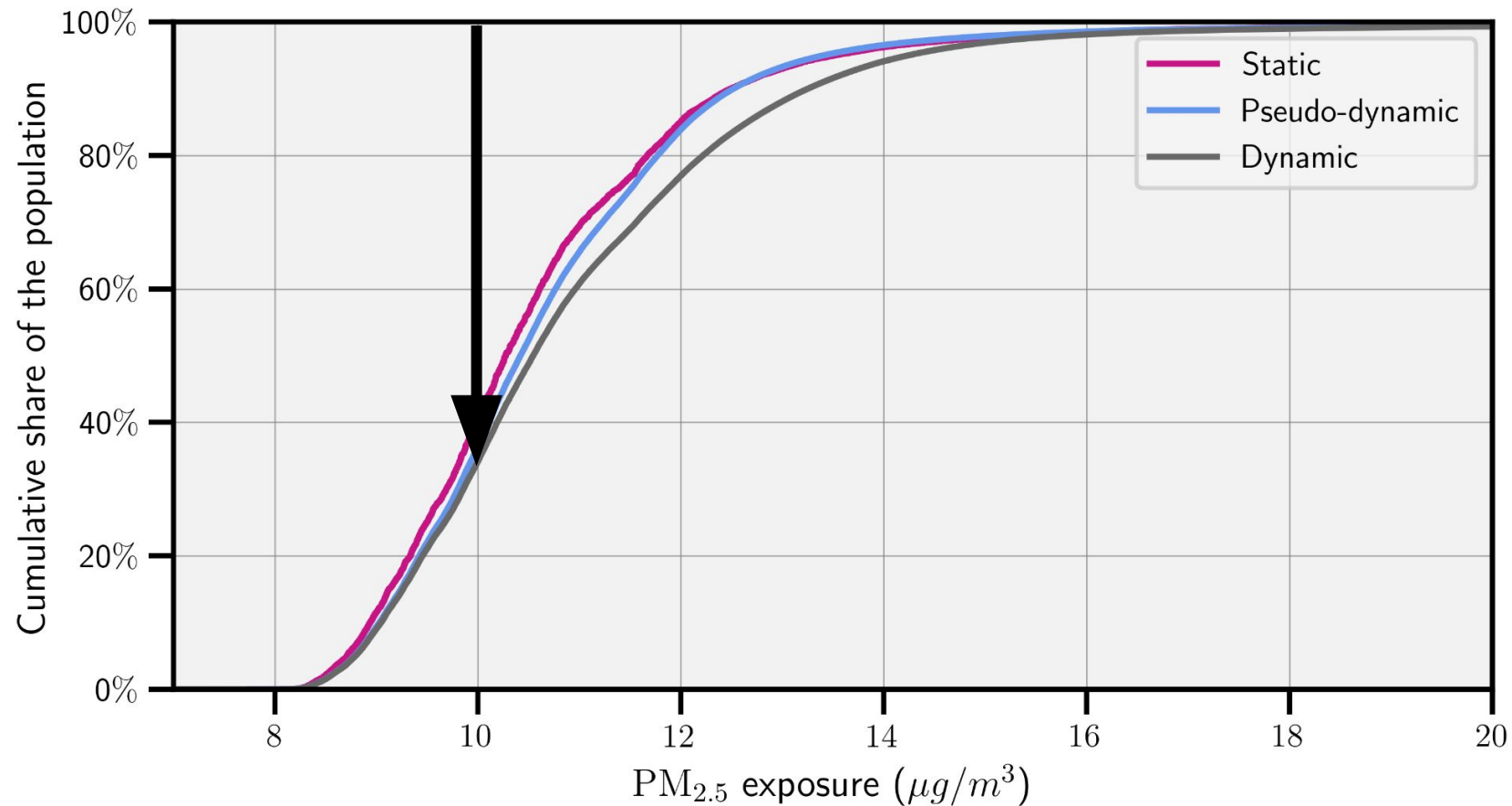
- **Cumulative distributions** of individual exposure by scenario
- **Mean relative difference** with the reference scenario (the most detailed one)
- **Maps** of mean relative differences **by area of residence**

The **most detailed approach** is believed to better estimate air pollution exposure, in absence of ground-truth data and is used as a **reference**.

How to read a comparison of scenarios with cumulative exposures



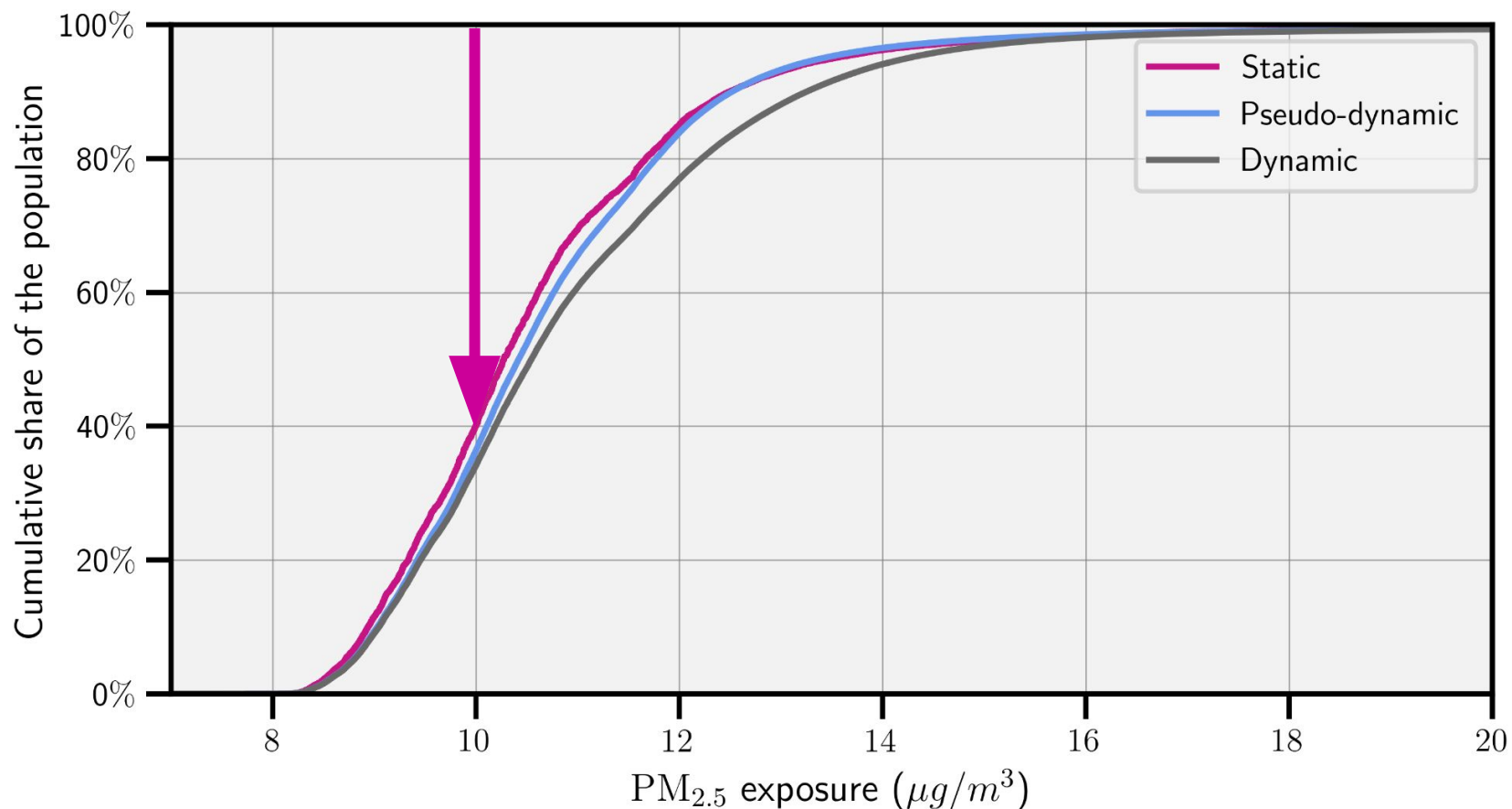
How to read a comparison of scenarios with cumulative exposures



Share of overexposed individuals

Dynamic scenario: **66%**

How to read a comparison of scenarios with cumulative exposures



Share of overexposed individuals

Dynamic scenario: 66%

Static scenario: 60%

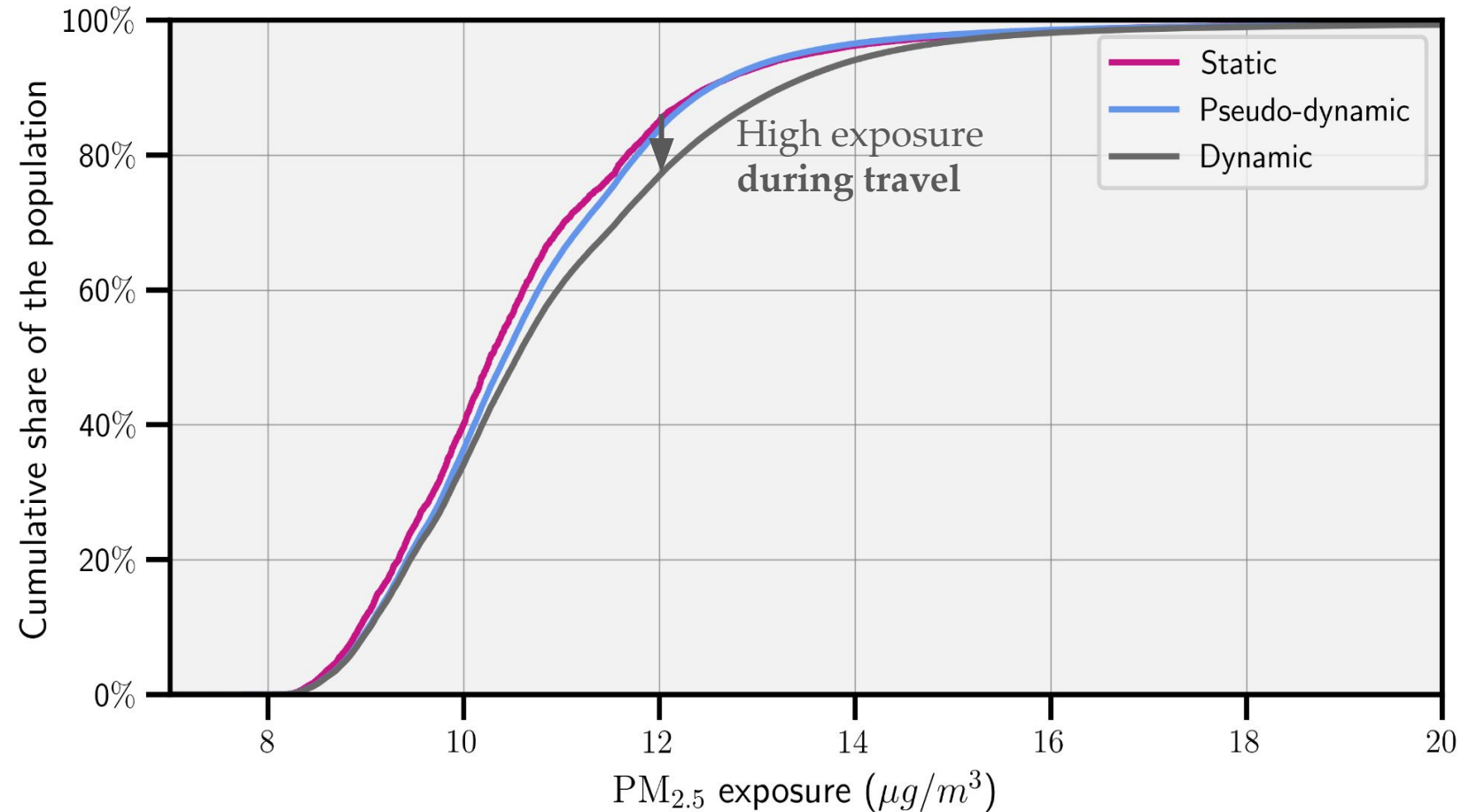


The lower the curves is, the more the population is exposed

The dynamic approach captures higher PM_{2.5} exposures

Average relative change compared with the dynamic scenario for PM_{2.5} exposure:

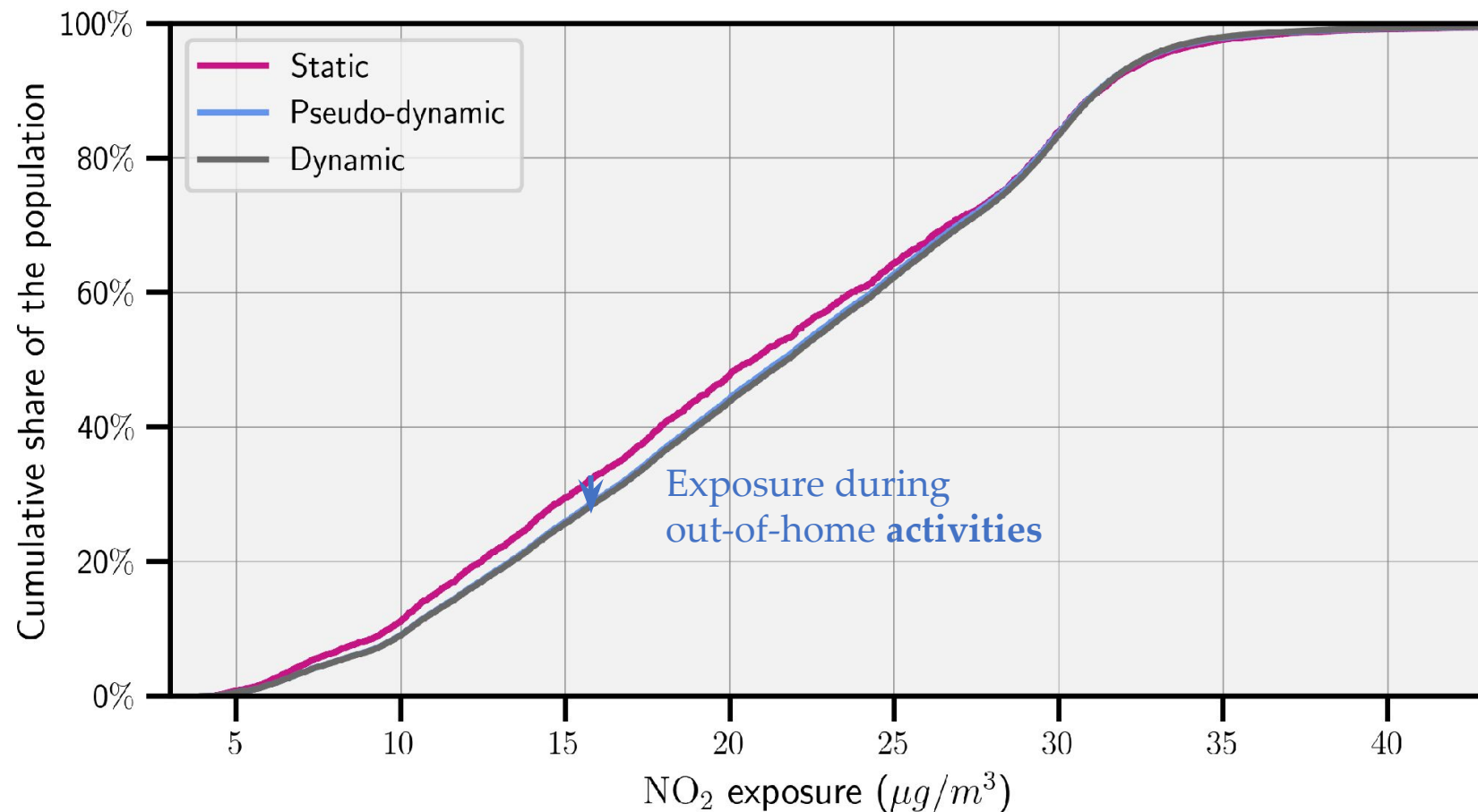
- - 2.5% for the static scenario with 9% standard deviation
- - 1.8% for the pseudo-dynamic scenario with 5% standard deviation



Results are more similar with pseudo-dynamic and static NO₂ exposures

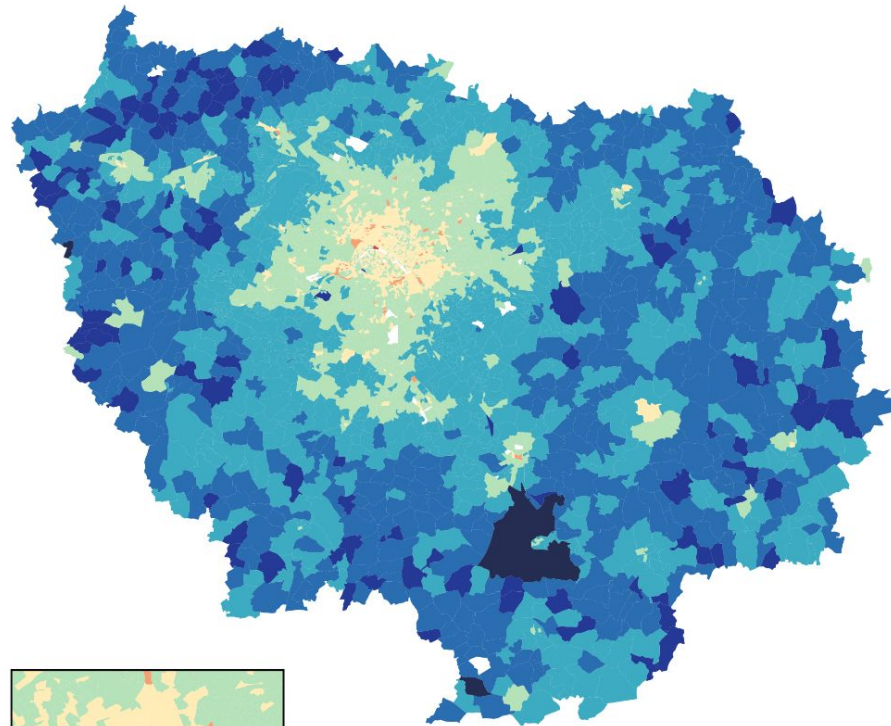
Average relative change compared with the dynamic scenario for NO₂ exposure:

- - 2.9% for the static scenario with **12% standard deviation**
- - 0.3% for the pseudo-dynamic scenario with 3% standard deviation



Static approach underestimates NO₂ exposure in the outer suburbs

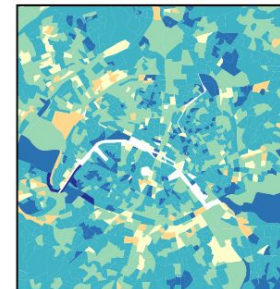
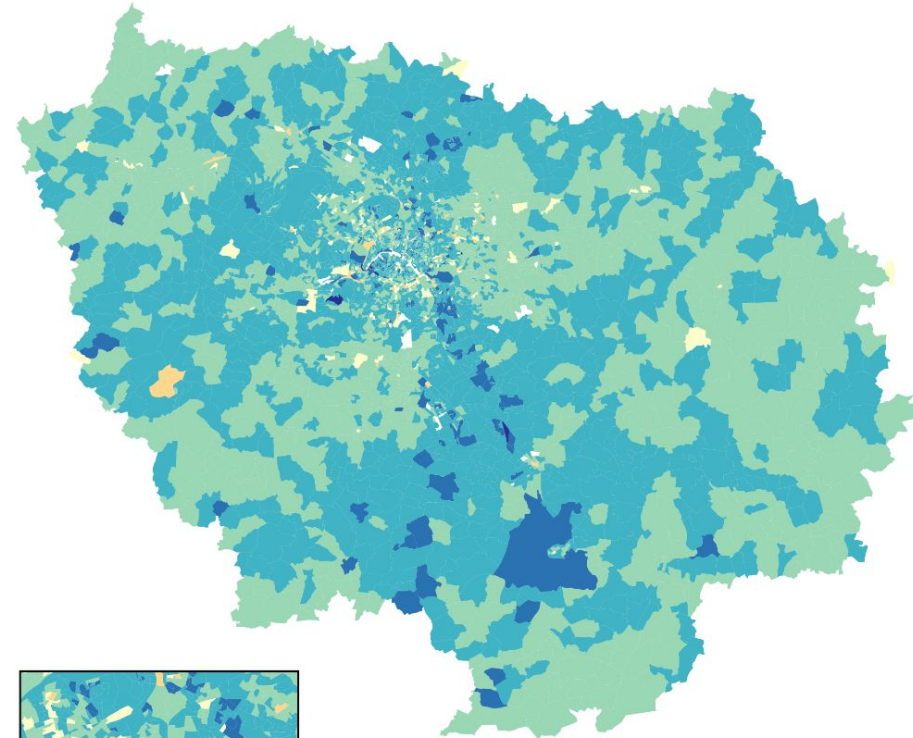
Static (vs dynamic) scenario
for NO₂ exposure



Mean relative change



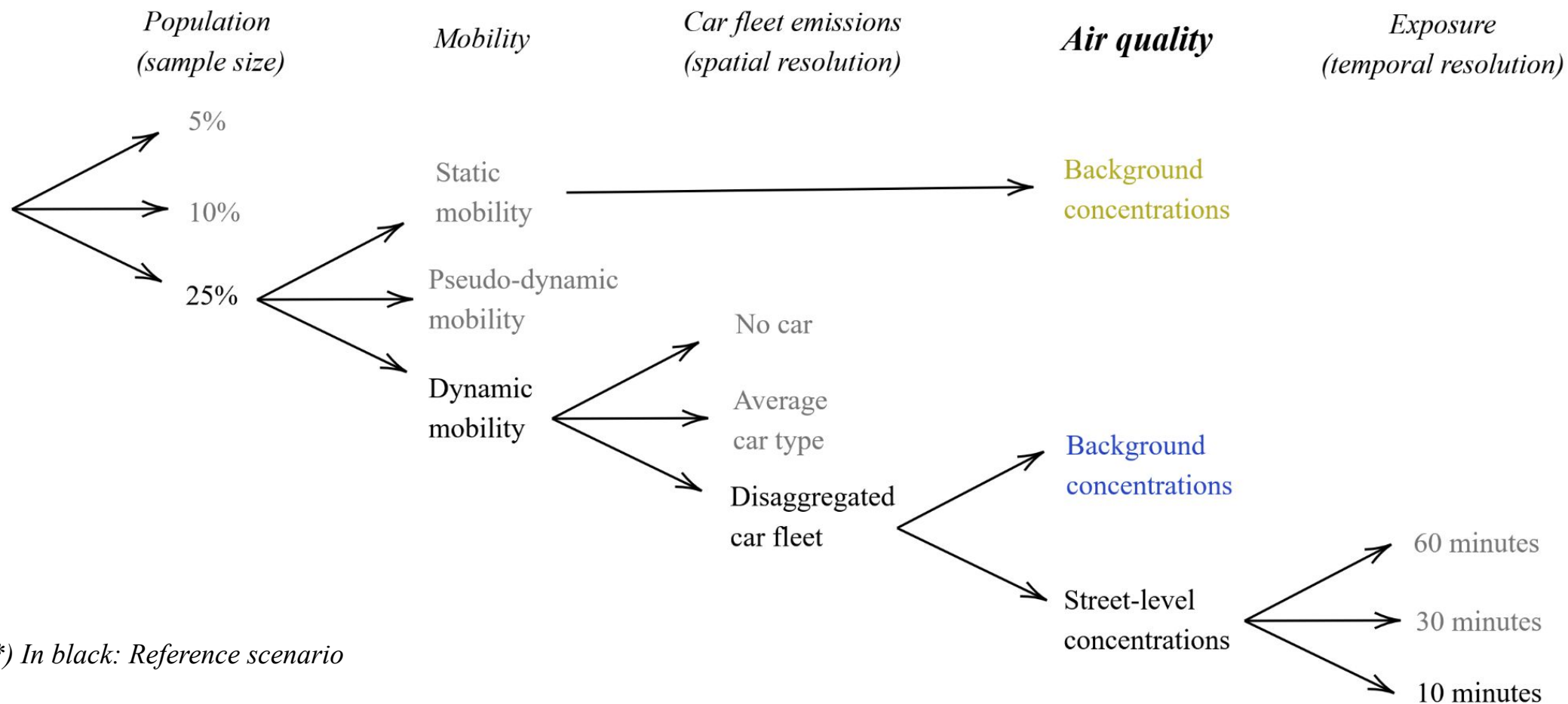
Static (vs dynamic) scenario
for PM_{2.5} exposure



Mean relative change



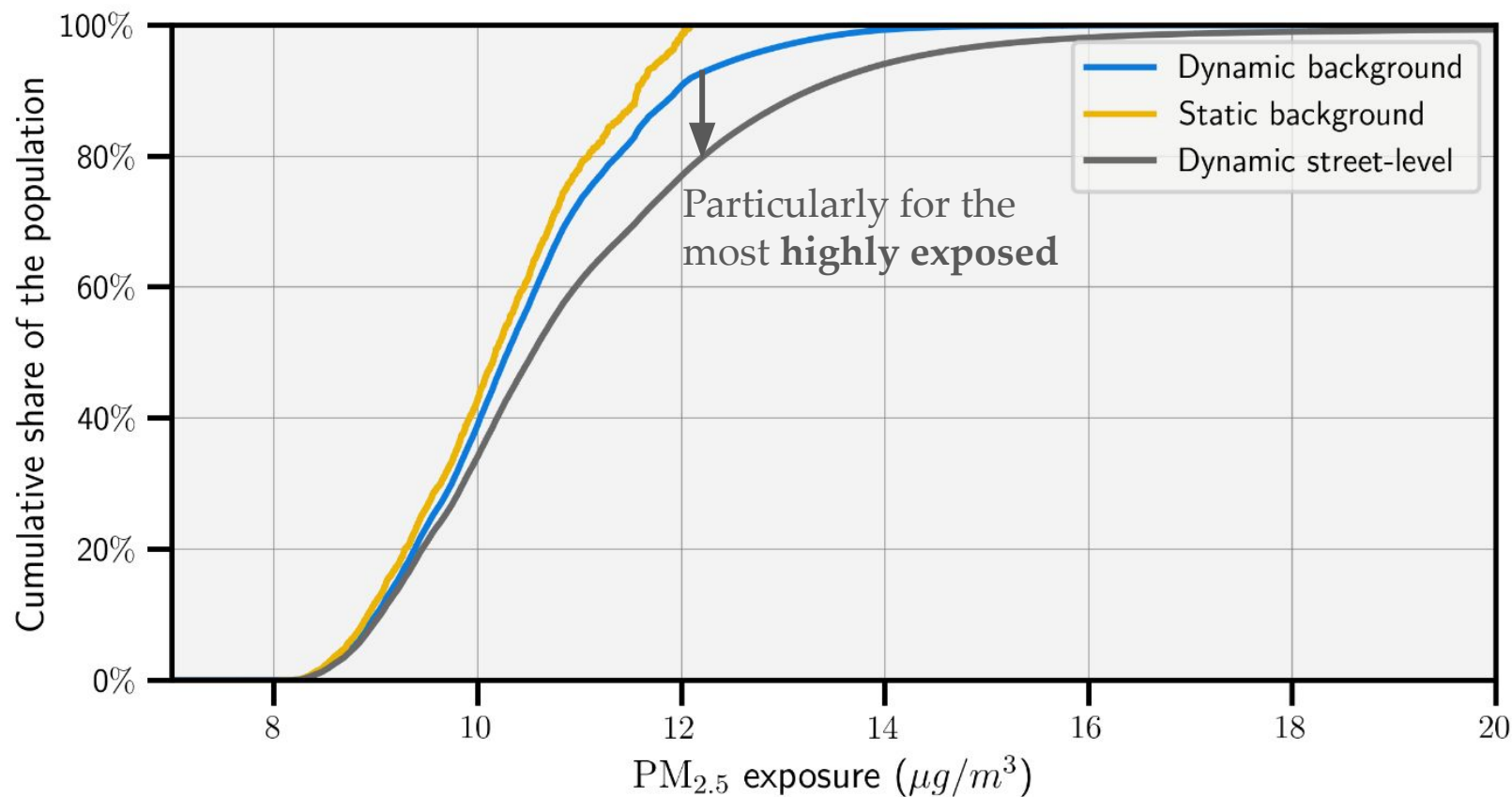
Air quality scenarios



Not representing canyon streets leads to underestimating exposure...

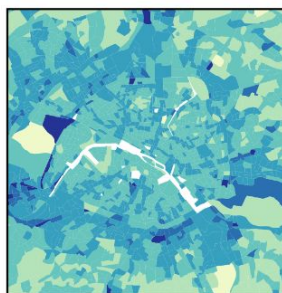
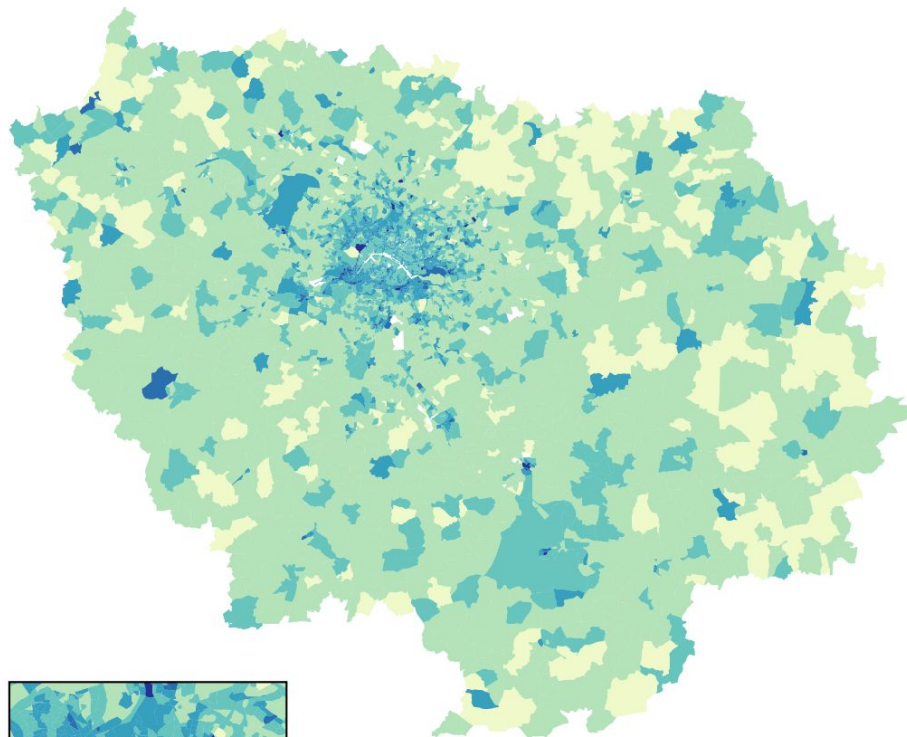
Average relative change compared with the reference scenario for $\text{PM}_{2.5}$ exposure:

- - 4.0% for the dynamic background scenario with 8% **standard deviation**
- - 5.6% for the static background scenario with 10% **standard deviation**.

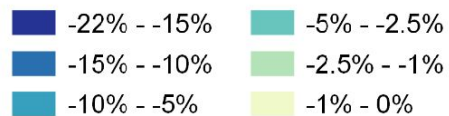


... particularly for those living in Paris and near the ring road

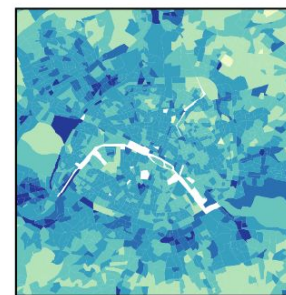
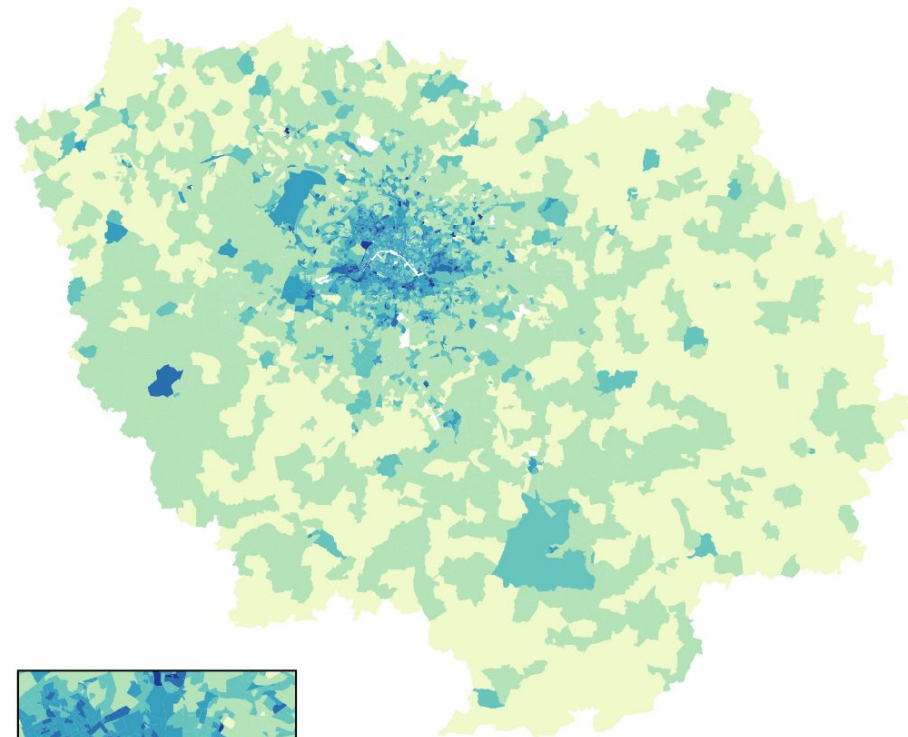
Background (vs street-scale) scenario
for NO₂ exposure



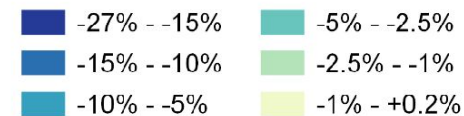
Mean relative change



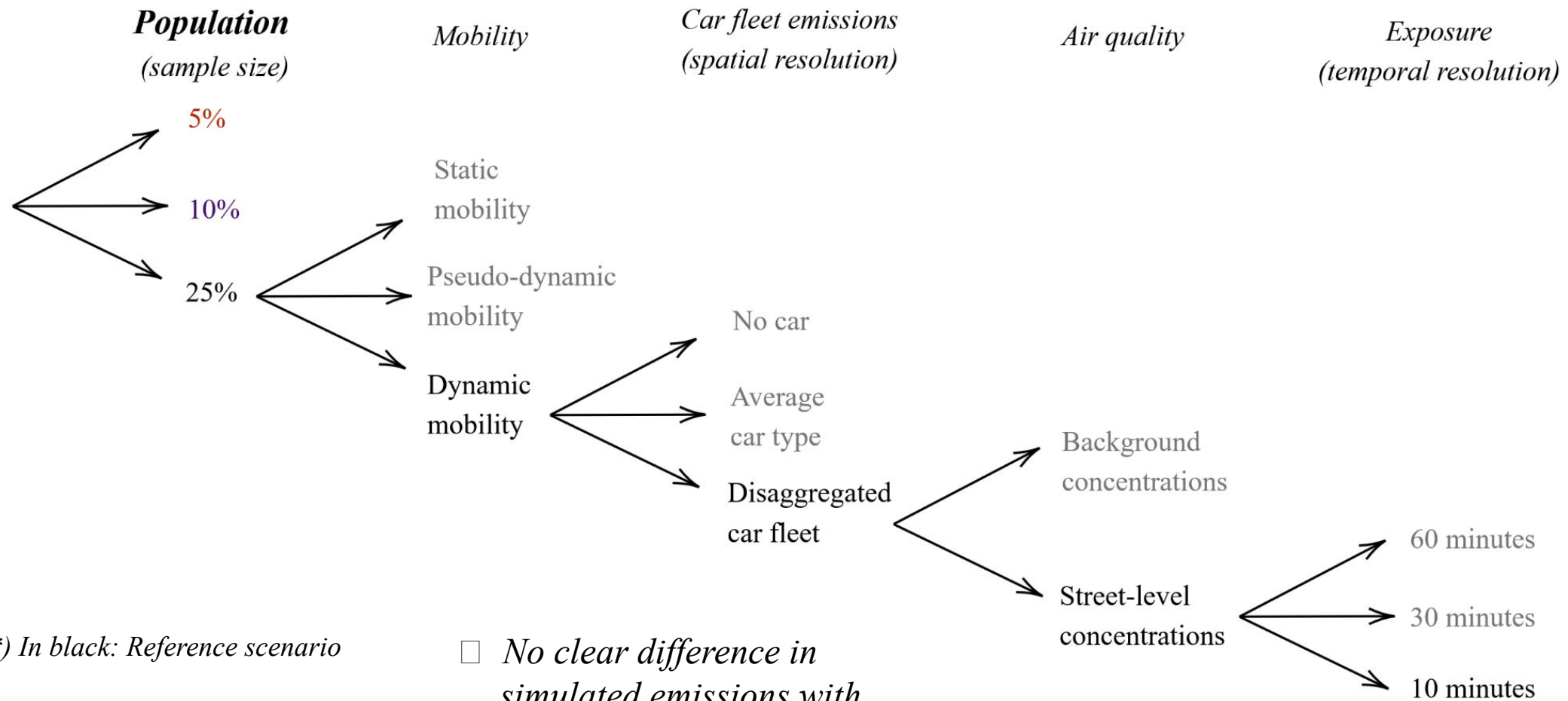
Background (vs street-scale) scenario
for PM_{2.5} exposure



Mean relative change



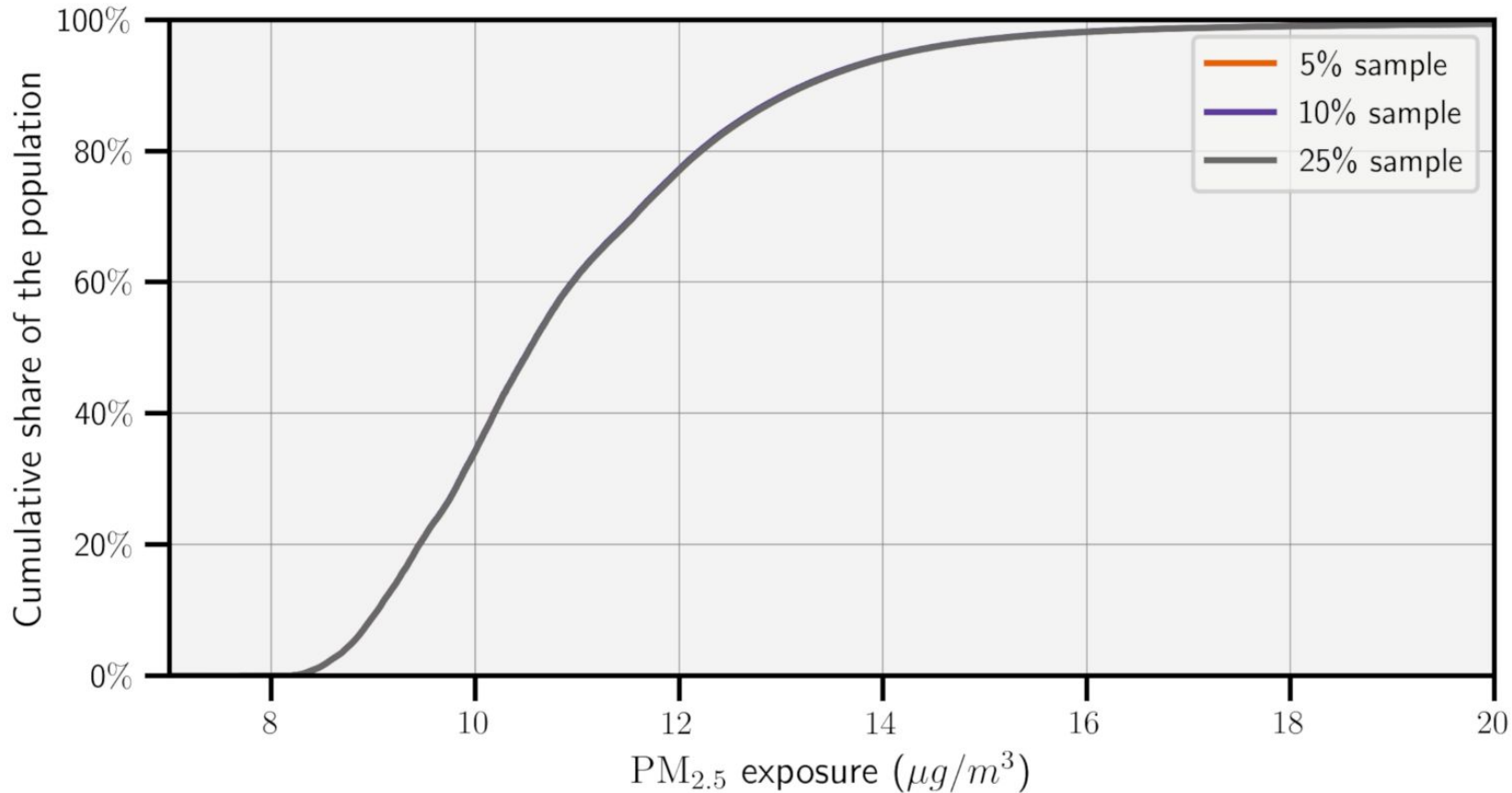
Population sample size



(*) In black: Reference scenario

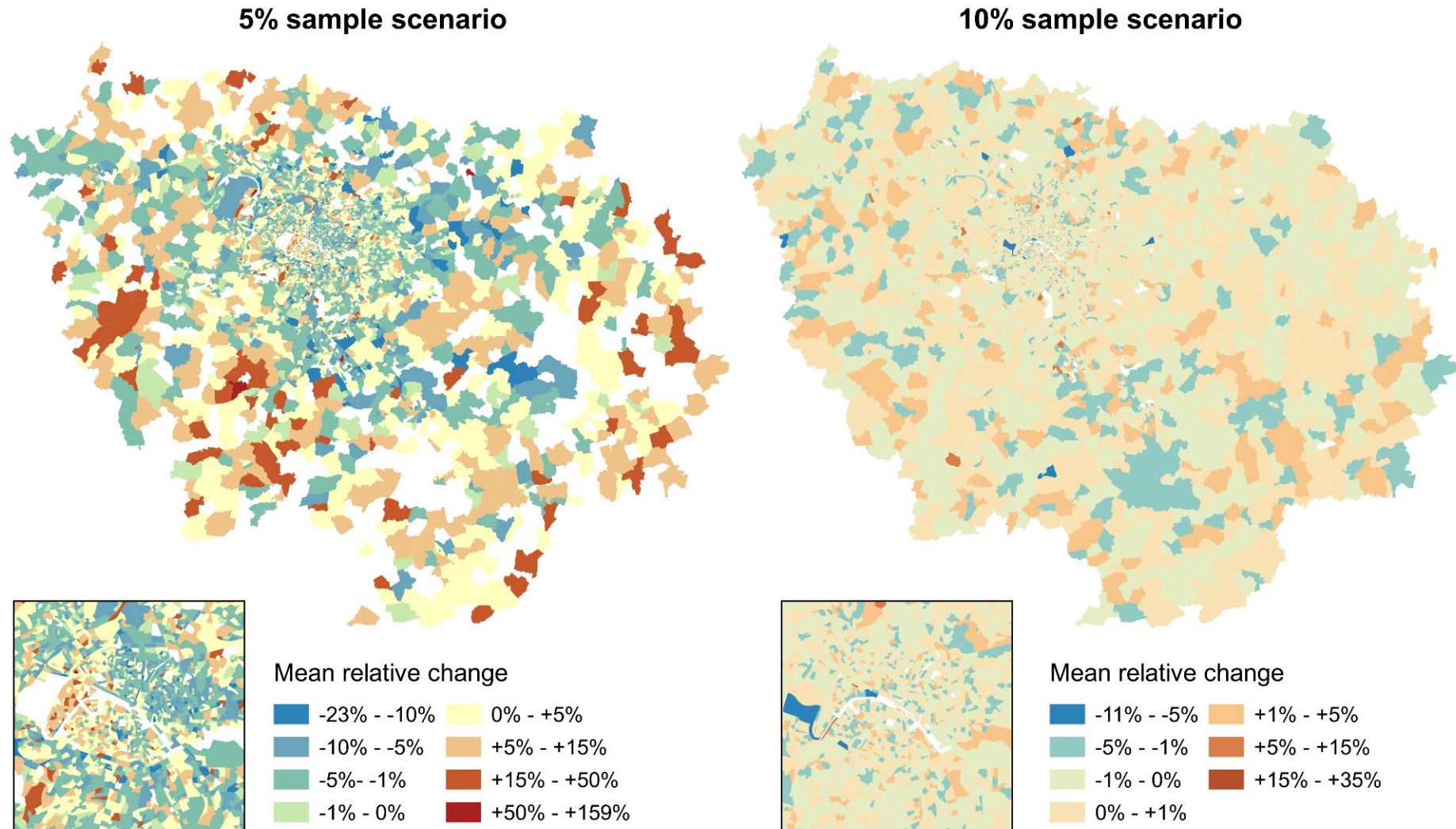
□ No clear difference in simulated emissions with different sample sizes

No effect of the sample size on cumulated exposure at regional scale...



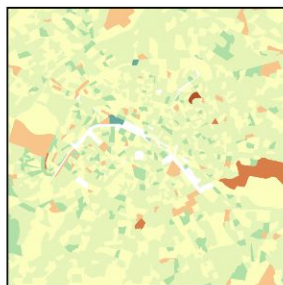
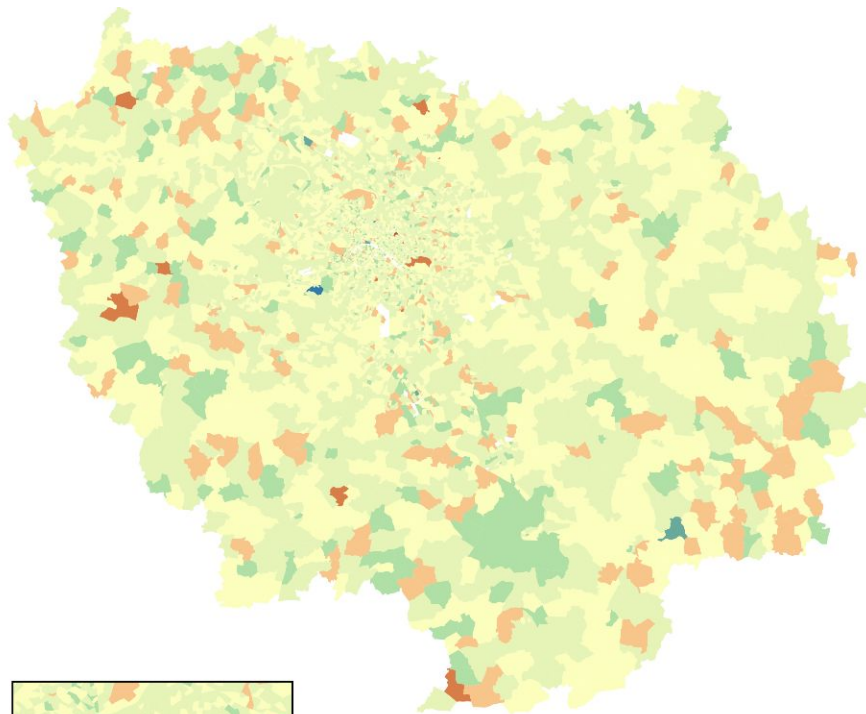
+ result **stable** with the repetition of the experiment at the regional scale

... But an important effect at the local scale with a 5% sample

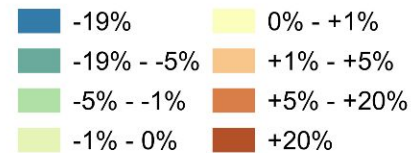


... And a low stochasticity at the local scale with the 25% sample

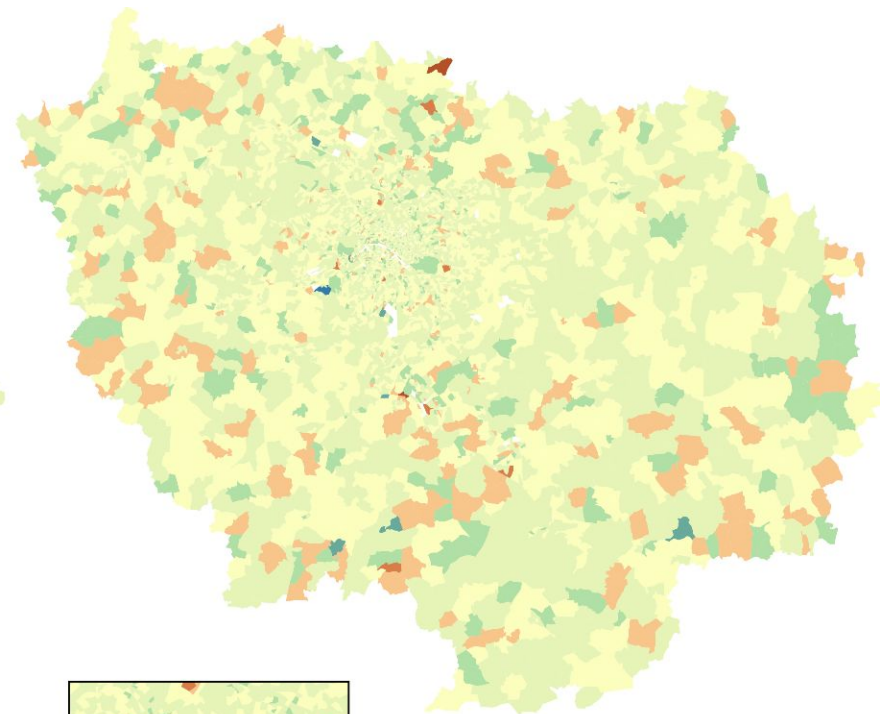
25% sample - simulation 2



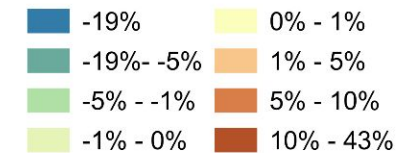
Mean relative change



25% sample - simulation 3



Mean relative change



Is model complexity worth it?

	Regional scale	Local scale / subpopulation
Mobility	+	++
Air quality in canyon streets	+++	+++
Population sample size	0	++++
Emissions by car type	+	+
Temporal resolution	++	++

Limitations and perspectives

- ❑ **Seasonal** exposure (in winter) to $\text{PM}_{2.5}$ and NO_2
- ❑ Limited representation of **microenvironments** (built environments, undergrounds, active mobility...)
 - ❑ Integrating indoor/outdoor infiltration ratios
- ❑ **Interday heterogeneity** of mobility and exposure
- ❑ **Computational time** ❑ to address for quantitative approaches
- ❑ Absence of **ground-truth data** for exposure
 - ❑ Cross subpopulation exposure modeling with portable sensor data



Thank you for your attention!

Contact: marjolaine.lannes@enpc.fr

Which hypotheses are not relevant to analyze exposure by age class?

Relative change in PM_{2.5} exposure by age range

	0-18 y.o.	19-30 y.o.	31-50 y.o.	51-65 y.o.	> 65 y.o.
Static	-2.5%	0.6%	-0.6%	-1.8%	-1.1%
Pseudo-dynamic	-2.7%	2.0%	1.2%	-0.2%	-1.2%
Background	-6.0%	-0.8%	-2.5%	-3.8%	-4.8%
Average car	-0.9%	7.2%	5.8%	3.5%	1.2%
No car	-4.7%	1.9%	0.4%	-1.4%	-3.1%
60 minutes	-10.1%	-4.8%	-6.4%	-7.8%	-9.0%
30 minutes	-3.7%	4.1%	2.7%	0.5%	-1.7%
5% sample	-2.0%	5.6%	4.0%	2.0%	-0.1%
10% sample	0.2%	3.3%	2.6%	1.6%	2.3%

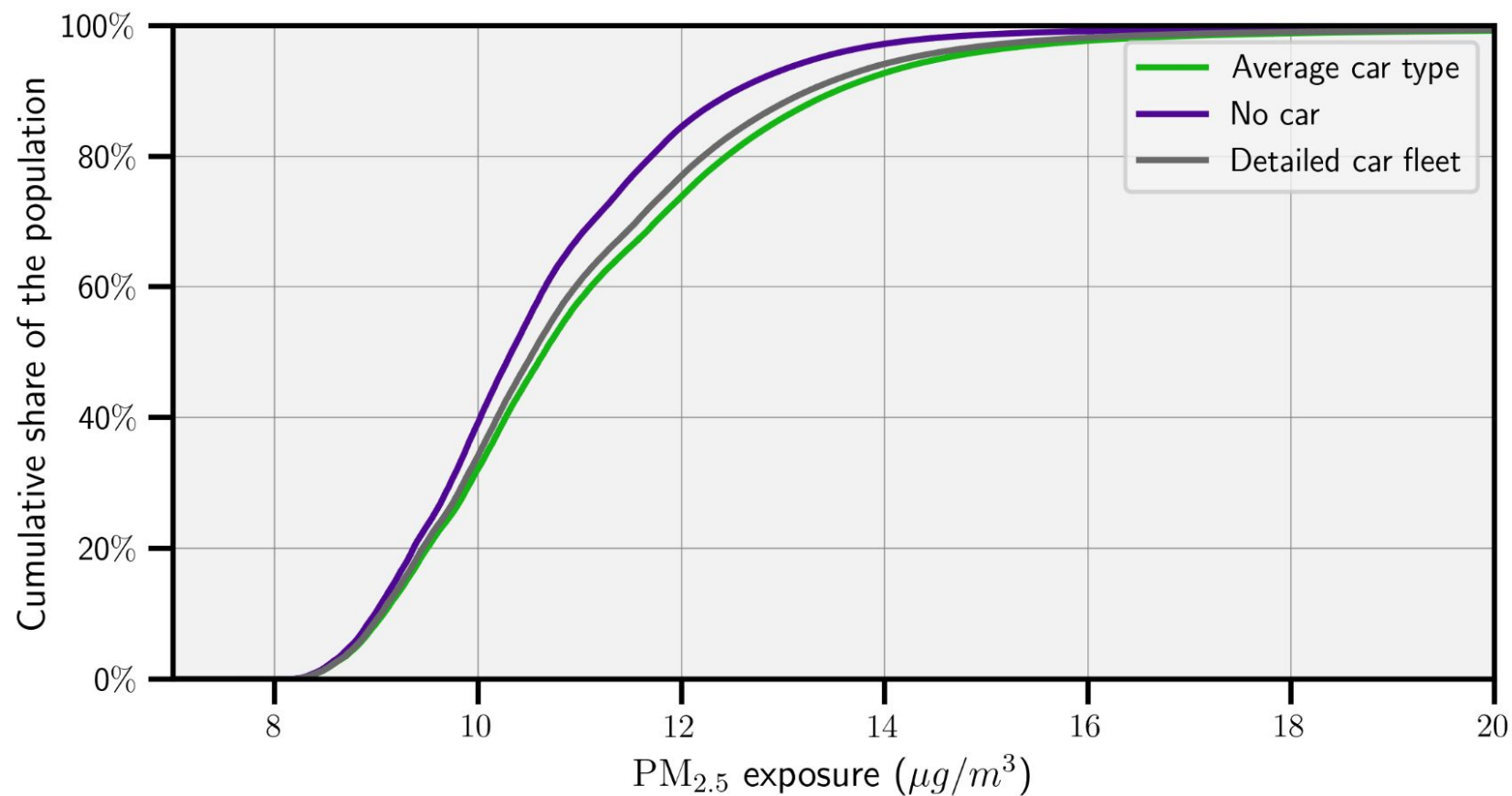
Underestimation of **children exposure**
 □ with background concentrations
 and 60 minutes resolution

Disparities in relative changes by age
 group with a time resolution of 30 min
 □ **10-minutes resolution** for assessing
 exposure by age group

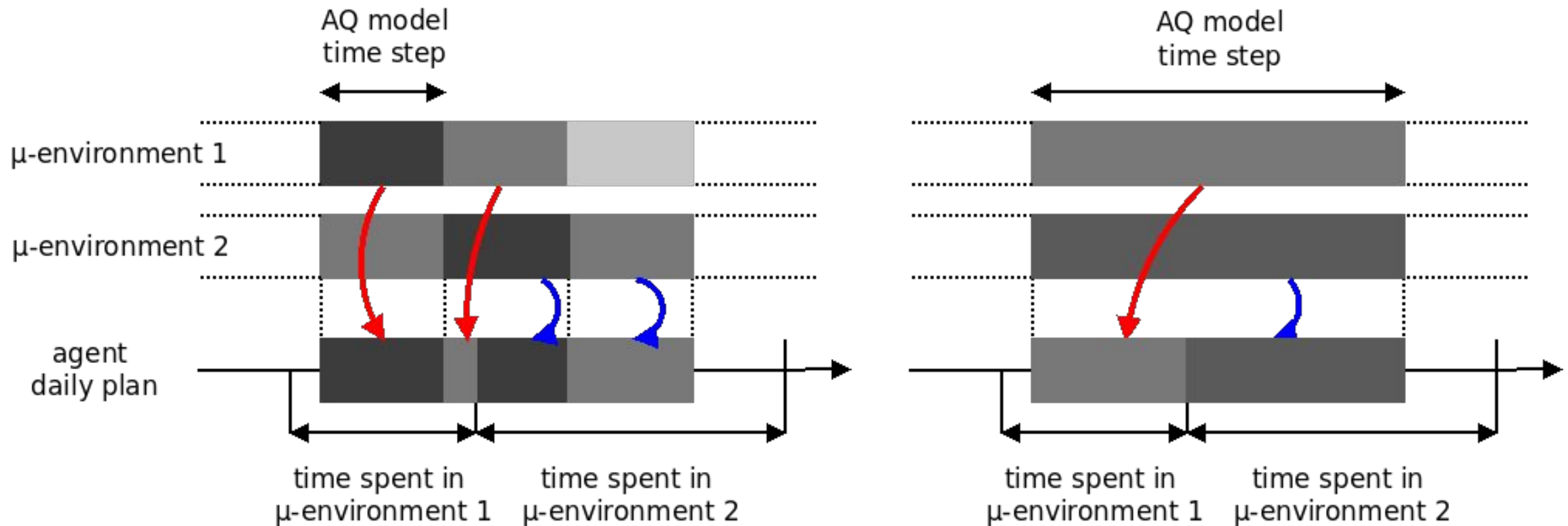
Representing car types has a slight impact on population exposure

Average relative change compared with the reference scenario for $\text{PM}_{2.5}$ exposure:

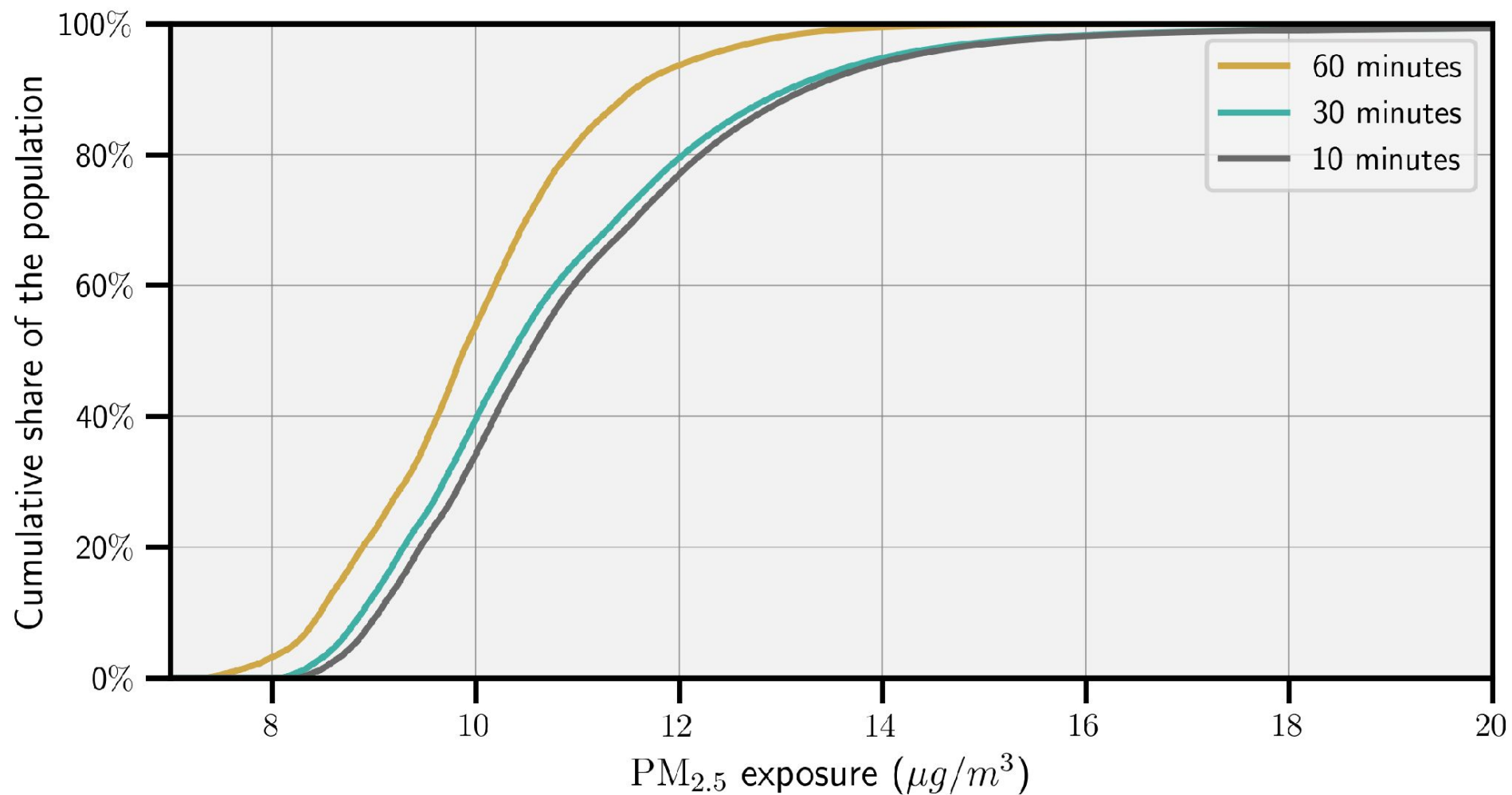
- + 1.2% for the emissions computed with an average car type, with 1% standard deviation
- - 2.8% for the emissions computed without cars



Model uncertainty: temporal resolution



Model uncertainty: temporal resolution

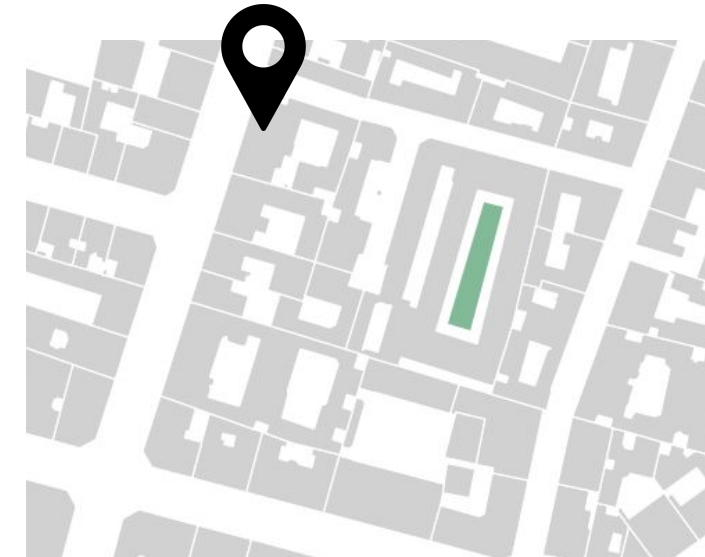
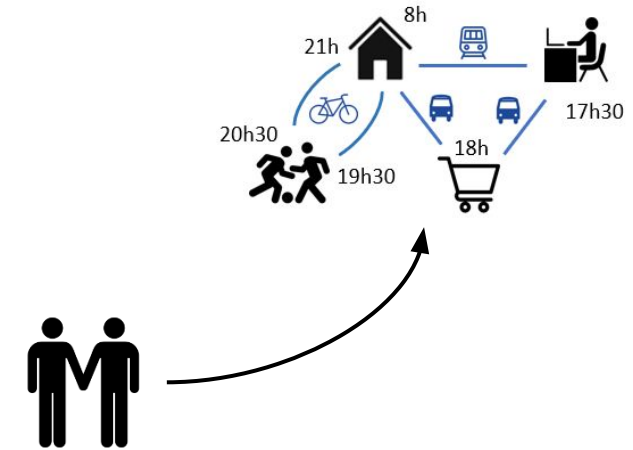


Population and travel demand

eqasim pipeline

- **Simulated** population for a **sample size** (here 25% i.e. ~3 million inhabitants of Île-de-France)
- Individuals structured in **households**, with **activity plans** and home/activity **locations**
- Input data:
 - **population census** (demographic data),
 - **tax data** (income),
 - buildings and company **locations**,
 - **household travel survey** (individual activity plans)

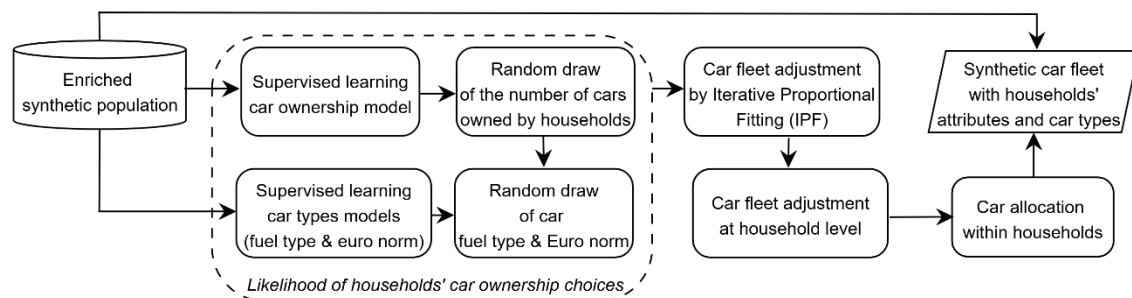
- Size: 2
- Gender: W - W
- Ages: 41 - 47
- Income: 4,000€



Air pollution: emissions and chemistry-transport models

Models	Model type	Spatial resolution	Temporal resolution	Chemistry
Road traffic emissions (HBEFA [5])	Traffic-state model	Road section Average or detailed car types	~ minute	Primary emissions for aggregated species (PM, COV, etc)
Background concentrations (POLAIR-3D [6])	Eulerian Chemistry-Transport Model	Grid (1 km resolution)	10 minutes	Includes secondary chemical generation of pollution
Concentrations in canyon streets (MUNICH [7])	Street box model	Canyon street	10 minutes	Includes secondary chemical generation of pollution

Car fleet synthesis: detailed vs average car fleet

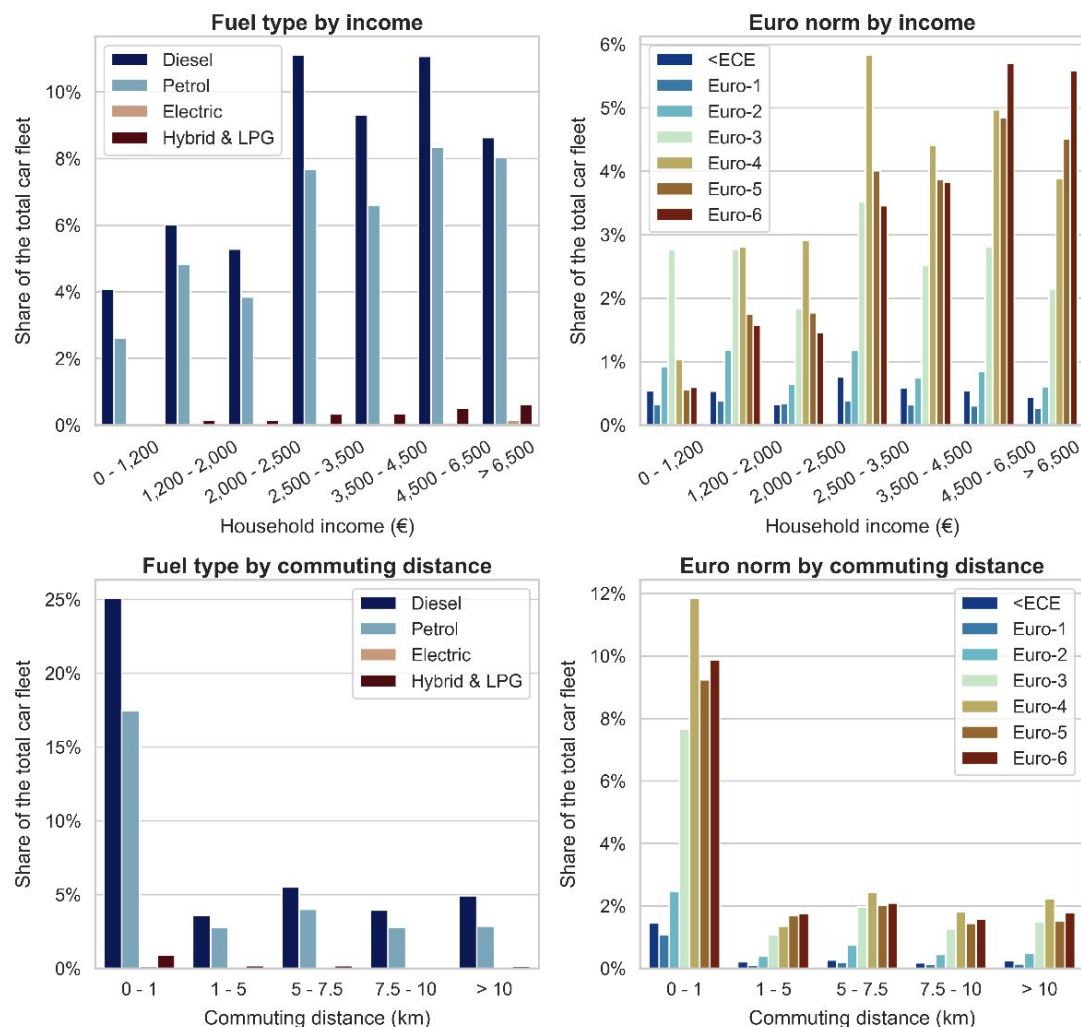


DETAILED CAR FLEET

Method to represent car types: fuel and age (Euro norm)

- allocated to households according to their socioeconomic and mobility-related characteristics
- matching aggregated statistics at a fine spatial scale (here municipality, Crit' Air LEZ sticker data)

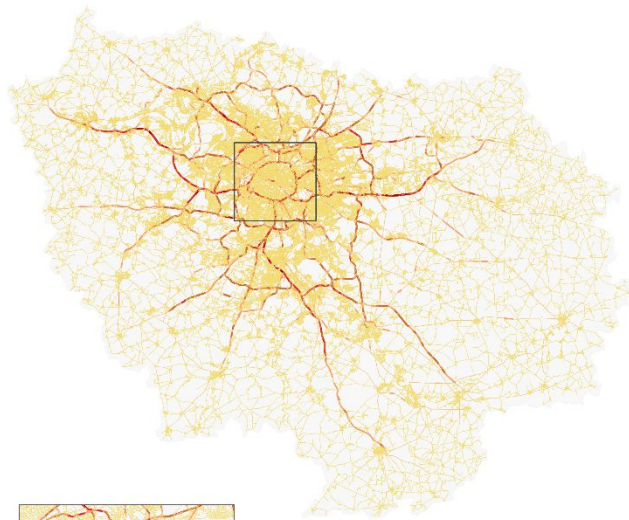
Distribution of car types by income and commuting distance in the adjusted car fleet



Emissions: detailed vs average car types

Place of emissions

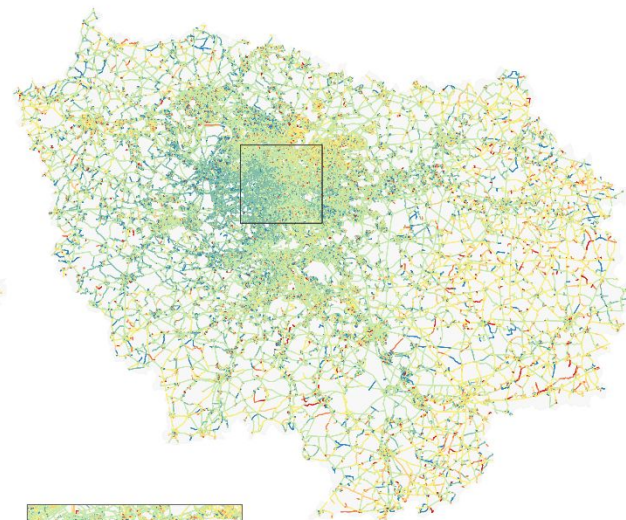
Nitrogen dioxide (NO₂) emissions



Daily emissions by road section (g)



Relative change in NO₂ emissions

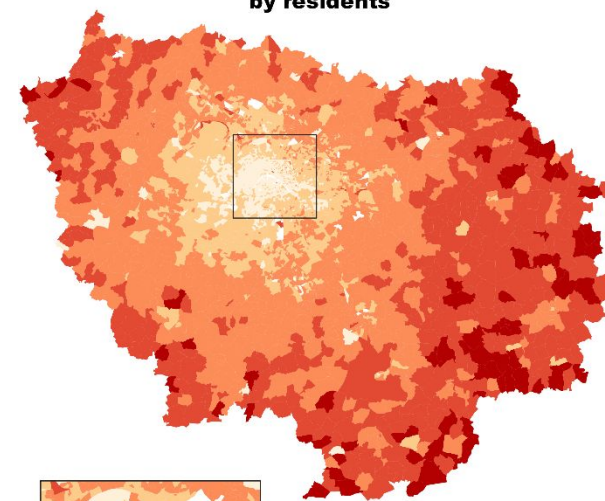


Relative change in emissions



Attribution of emissions

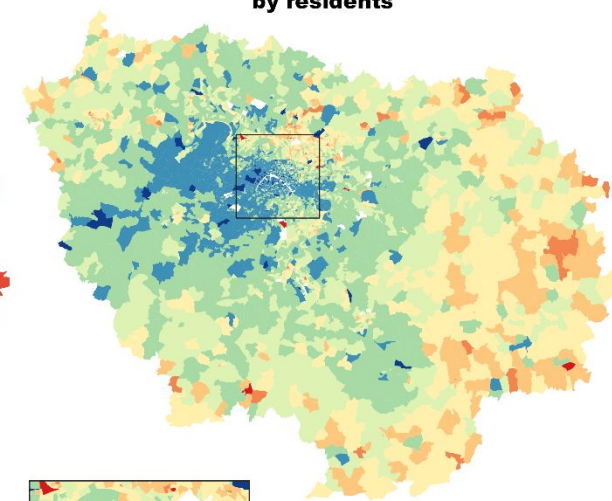
Daily individual emissions of NO₂ by residents



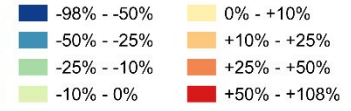
Mean individual emissions (g) of residents using their cars as drivers



Relative change in NO₂ emissions by residents



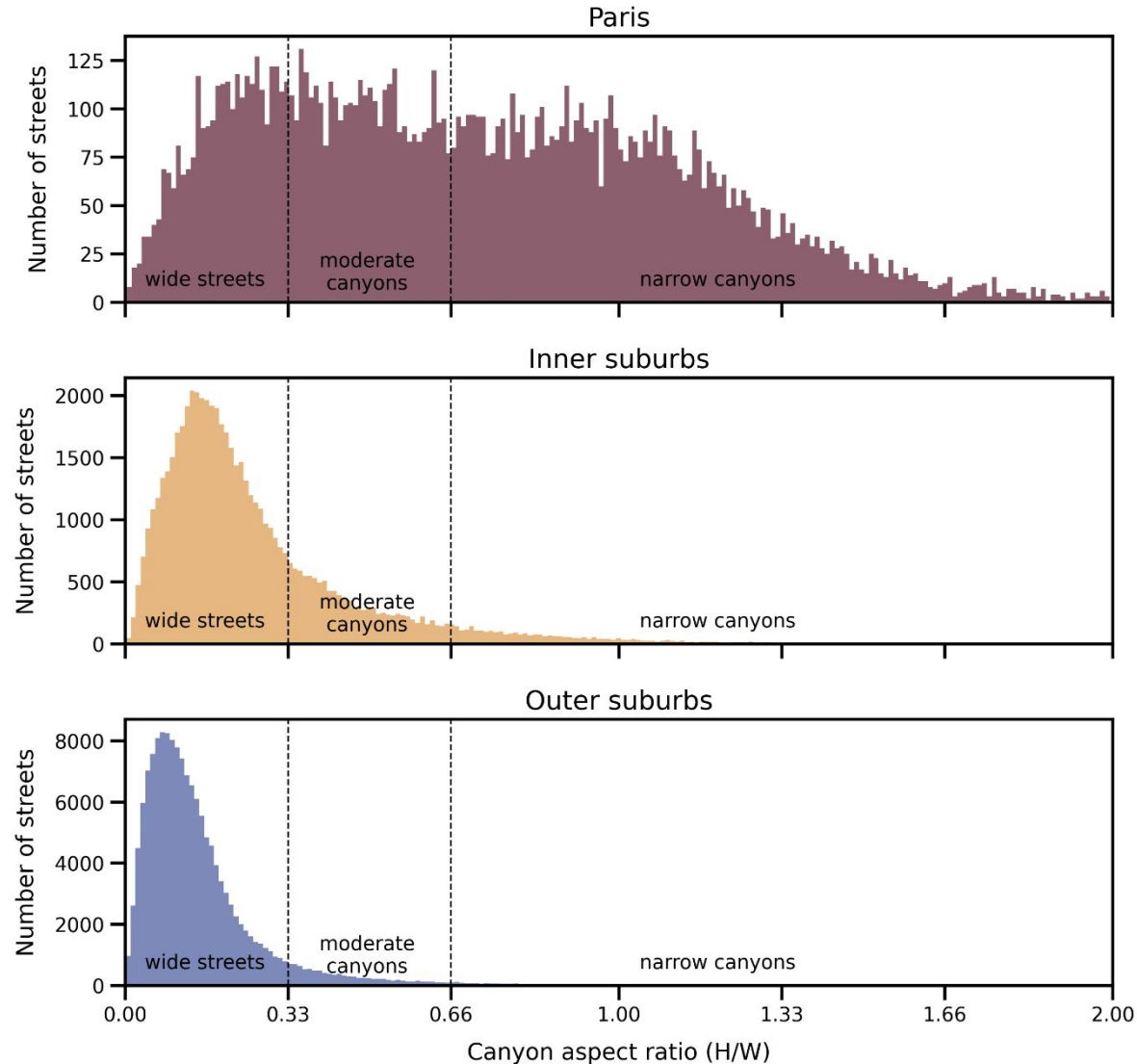
Relative change in emissions of the detailed cars compared to average ones



NO₂: -13.8% of emissions at regional scale
PM_{2.5}: -2.5% of emissions at regional scale
BC: -10.9% of emissions at regional scale

Modeling the canyon street network

Results on the urban form



Canyon aspect ratio = height to width
□ criterion for canyon streets selection

Larger proportion of streets with $H/W > 1/3$
in **Paris** than the suburbs

Numerous *moderate canyons* ($2/3 > H/W > 1/3$)
in the **inner suburbs**